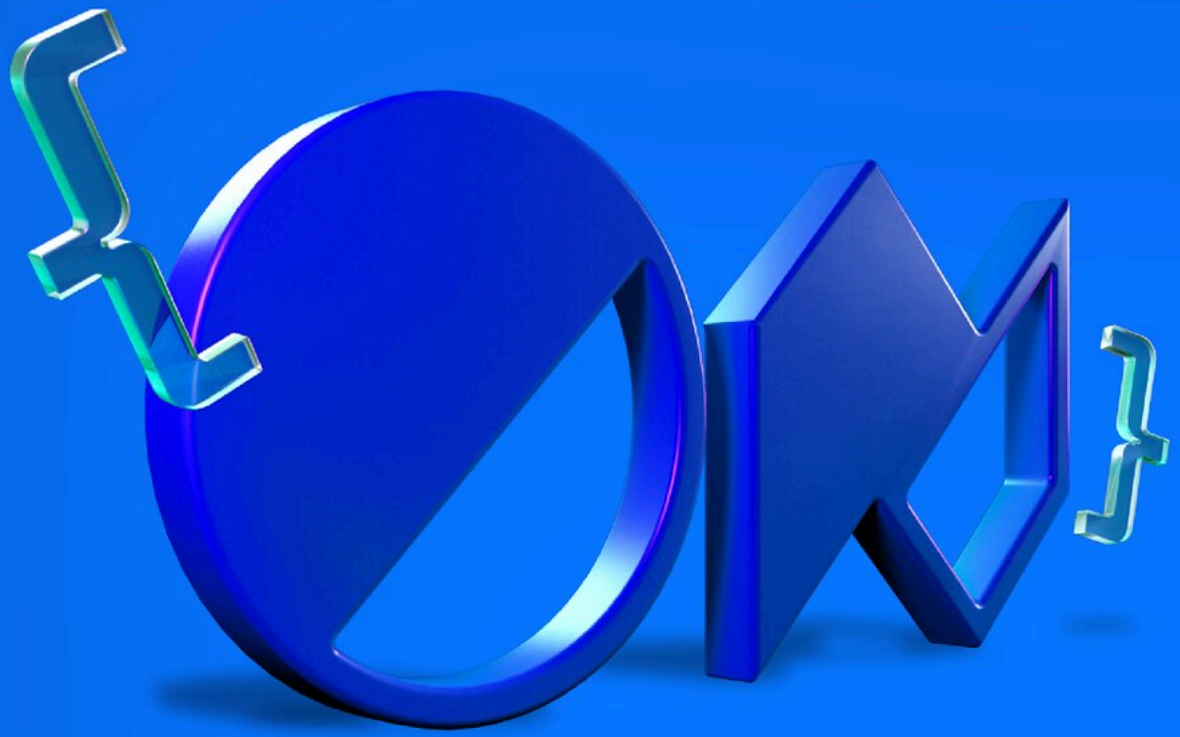
# Search is test-time compute

**Han Xiao**, VP of AI, Elastic

ELASTICON AI SAN FRANCISCO · NOVEMBER 4, 2026

SCAN FOR THE SLIDES

hanxiao.io/elasticon-sf-2026



elastic

# Search has mostly improved by training harder.

Encoders, data, indexes and rerankers all improve before the first query.



# Recent LLMs found a second axis: compute at inference

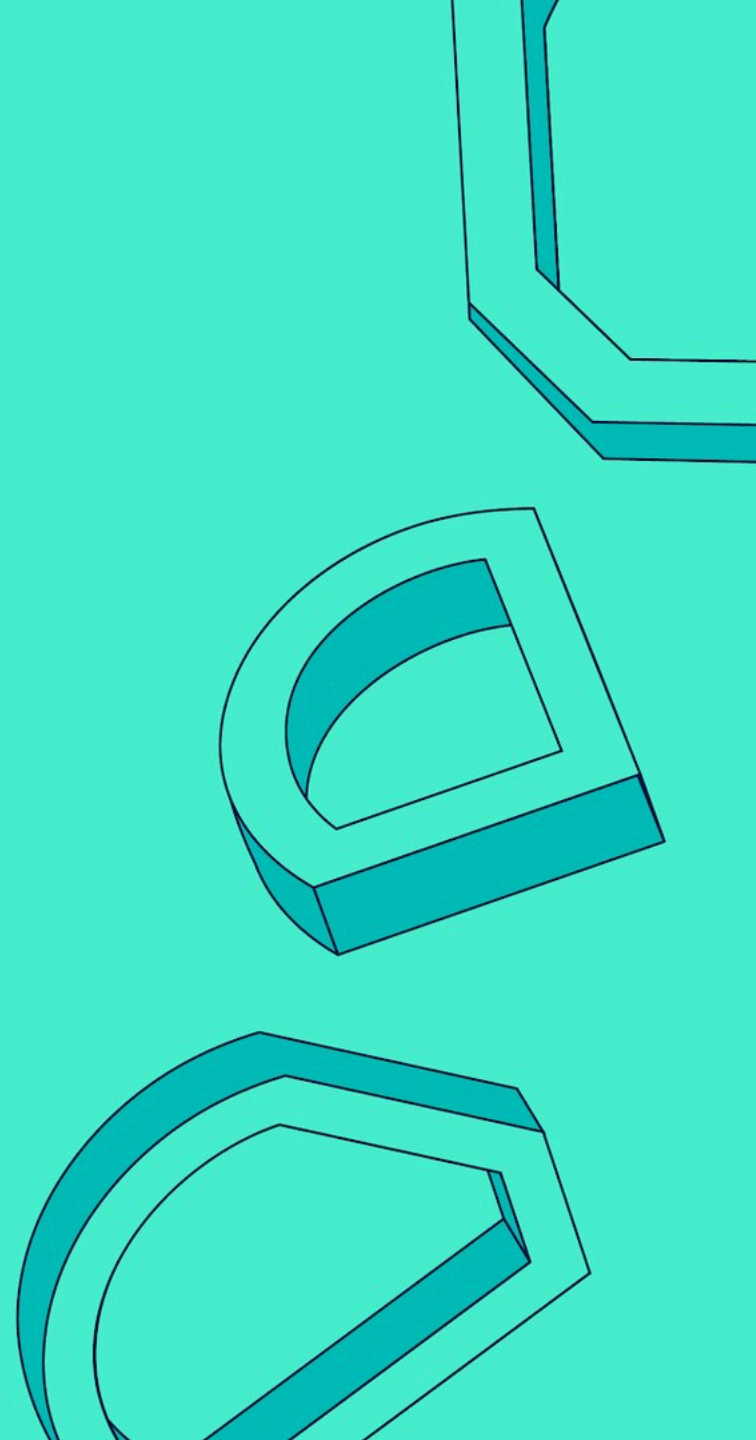
With test-time scaling, the same model does more work per query and quality rises with the budget.




> Having a bot think for just **20 seconds** in a hand of poker got the same performance boost as scaling up the model by **100,000x** and training it for 100,000 times longer.
>
> Noam Brown, OpenAI, 2024

# Fun fact: search has always lived on that axis.

Embedding, retrieval, reranking, expansion and fusion all run after the user presses Enter.



# Inference time is a currency, not a KPI

UNTIL NOW: A KPI

## Inference time is a KPI

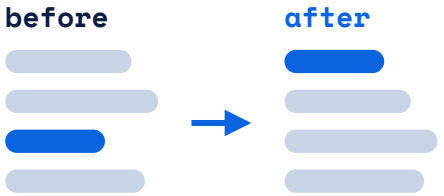
Latency is a limit written into the SLA. **Better accuracy requires a bigger model.**

BECOMES

IN THIS TALK: A CURRENCY

## Inference time is a currency

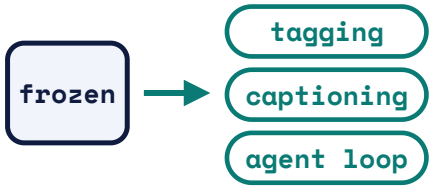
**Keep the weights and raise the budget from 10 ms to 100 ms. What does it buy?**




BUYS

### Relevance

Better ranking on the same task.



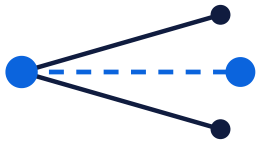

BUYS

### A new capability

Tasks the model was never trained for.

# Four training-free demos

Each one buys relevance or a new capability with compute at test time.



01

## Programs that transfer

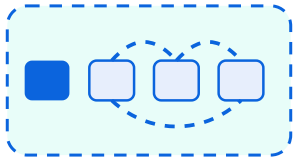
An LLM writes retrieval programs for a frozen encoder. The cheapest ones transfer to unseen encoder families.



02

## Open-vocabulary tags

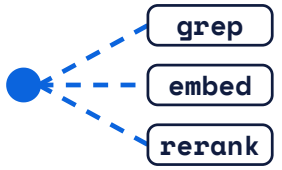
A frozen retrieval model becomes an image tagger.



03

## One call replaces an index

One listwise reranker call over a whole page replaces chunking, embedding and indexing.




04

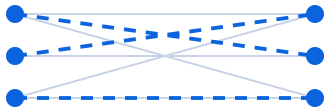
## Agent-built pipelines

For each query, an agent builds a search pipeline from local tools.

# Three ways to spend compute on a frozen model

All three keep the weights fixed and differ in where new information comes from.

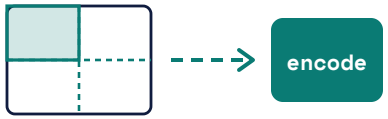
## A DEEPER PASS



Read every token or patch vector of the pass you already ran.

**New information: from inside the pass**

## B MORE PASSES




Run the encoder again on new views, such as sentences or image crops.

**New information: from the input**

## C CALIBRATION



Transform existing vectors with whitening, label propagation or optimal transport.

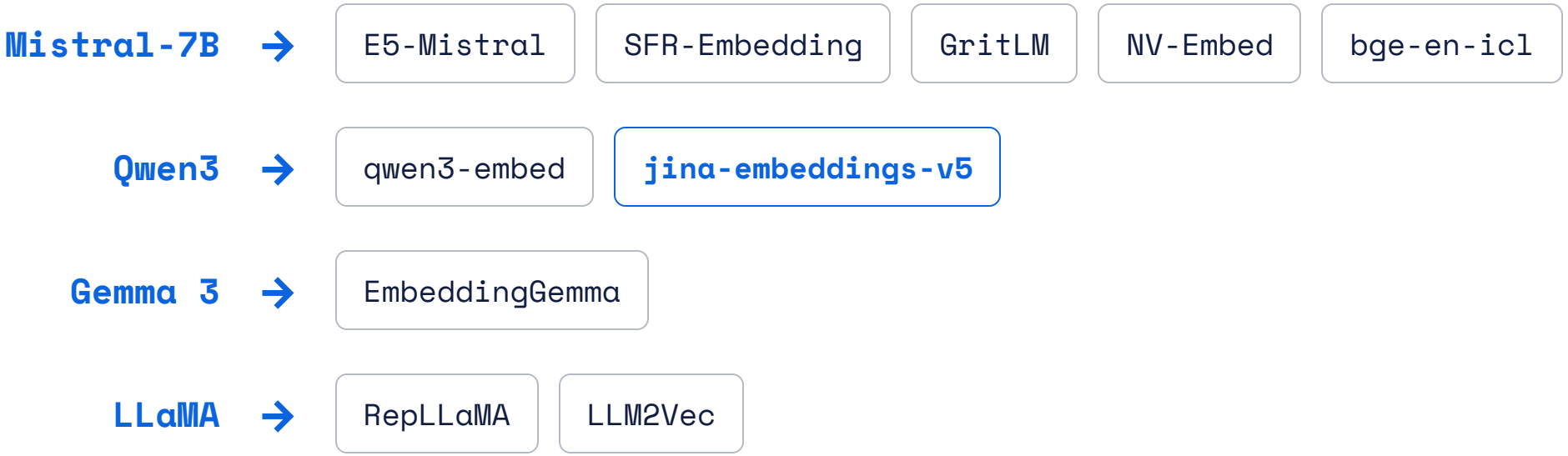
**New information: none**

01

# Programs on a frozen encoder

# Current small embedders are distilled from LLMs

If test-time compute works in the LLM representation space, distilled encoders may inherit it.




First raised in **jina.ai/news/scaling-test-time-compute-for-embedding-models**

# Scoring runs from one cosine to late interaction

A single-vector encoder can split a document into sentences and keep the best match.

### SINGLE-VECTOR COSINE

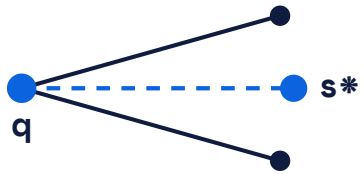



$$\cos(q, d)$$

one vector per document

**FROZEN BASELINE**

### SENTENCE MAXSIM

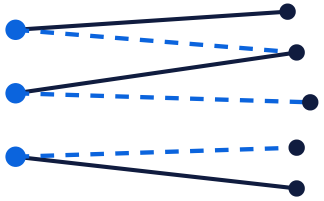



$$\max_s \ \cos(q, s)$$

same encoder, one vector per sentence

**TEST-TIME COMPUTE**

### COLBERT MULTI-VECTOR



$$\sum_i \max_j \ \cos(q_i, d_j)$$

max over document tokens for each query token

**NEEDS A MULTI-VECTOR MODEL**

**Rules:** only the frozen encoder in the query path, with no LLM, second retriever or learned parameters. One **embed_fn** call is one unit of compute.

# An LLM agent writes programs over the frozen encoder

Each round changes one program, evaluates it at a fixed budget and keeps it only if the metric improves.




| WRITES | ENCODER | SCORES | LOGS | KEEPS |
|---|---|---|---|---|
| one program per round | 239M frozen | 14 MMTEB tasks ΔnDCG@10 | Δ, cost, parent and lesson | 144 scored programs |

> The evaluator rejects hyperparameters, task-specific routing, external models and learned parameters. **arXiv:2605.11374**

# We search on one encoder and test on three others

DISCOVERY · THE LOOP SEARCHES HERE

**j-v5-nano**
239M · Jina
DISCOVERY ENCODER

**small and fast, so each round is quick**

SEARCHED OVER
**14 MMTEB Tier-1 tasks** · legal, financial, long-doc, general

the same program

HELD-OUT · THE LOOP NEVER SEES THESE

**j-v5-small**
568M · Jina
SAME FAMILY

**gemma-300m**
303M · Gemma 3
UNSEEN FAMILY

**qwen3-0.6b**
600M · Qwen3
UNSEEN FAMILY

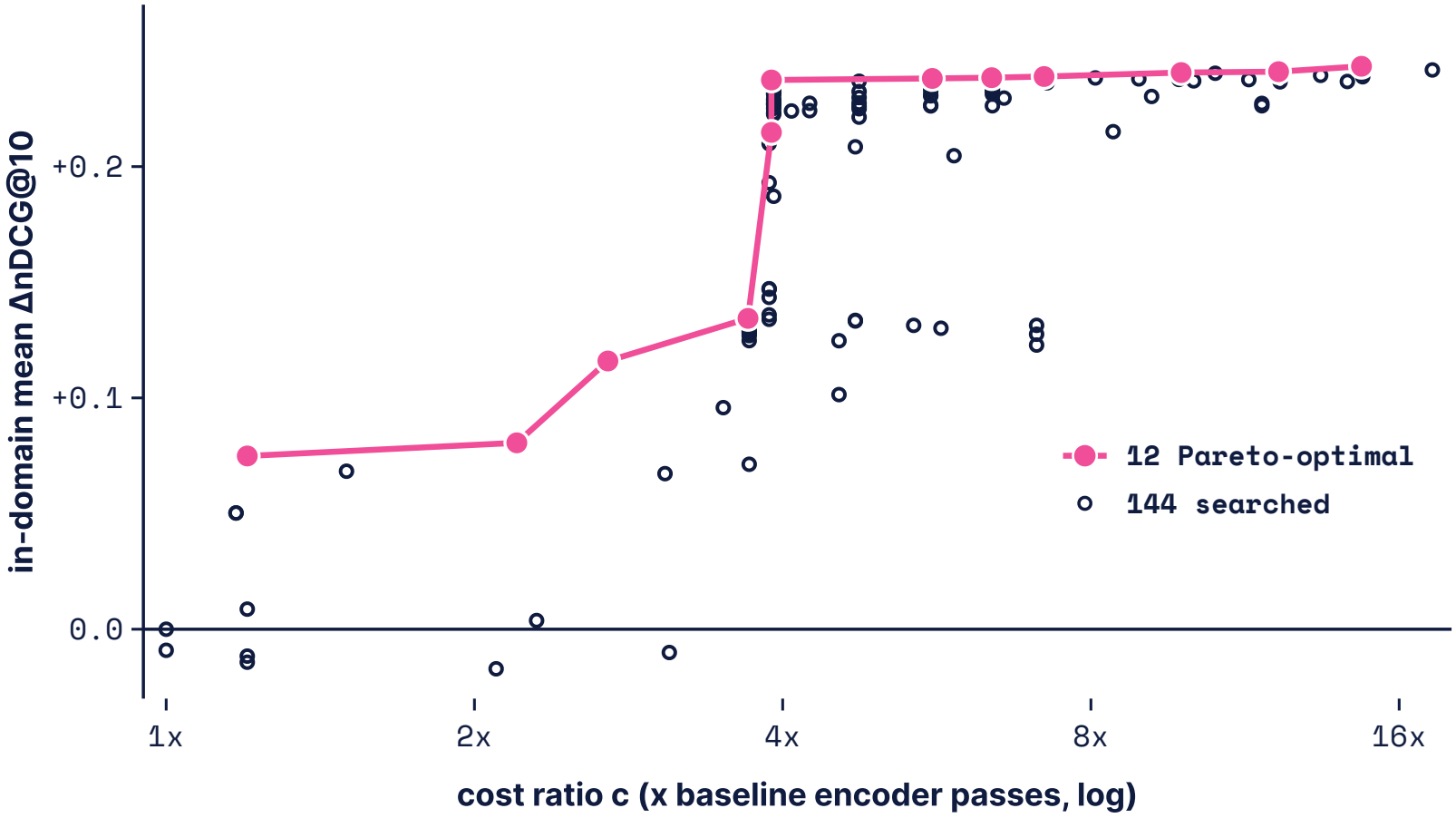
**unseen families test transfer**

TESTED ON
**19 MMTEB Tier-2/3 tasks** · summarization, QA, fact-check, …

**Metric:** ΔnDCG@10 against the cosine baseline. **Cost c:** encoder passes per query, relative to the baseline.

# Rewarding compute gives a Pareto curve

Over 144 programs, the in-domain mean ΔnDCG@10 climbs from +0.07 to +0.24.




**144**

programs searched

**12**

Pareto-optimal, from c = 1.2 to 14.7

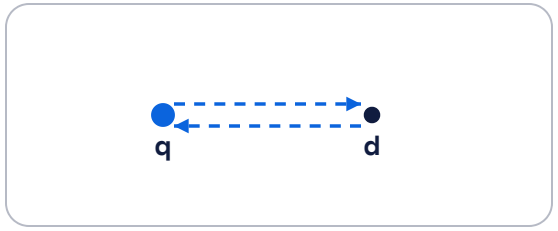
**+0.24**

best in-domain mean ΔnDCG@10

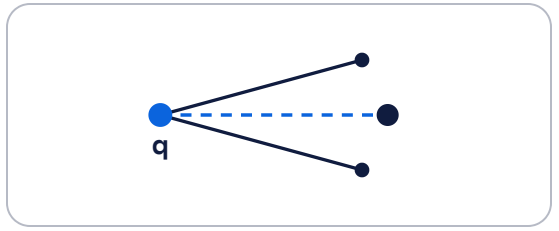
Measured on the 14 discovery tasks.

# Twelve programs on the compute frontier

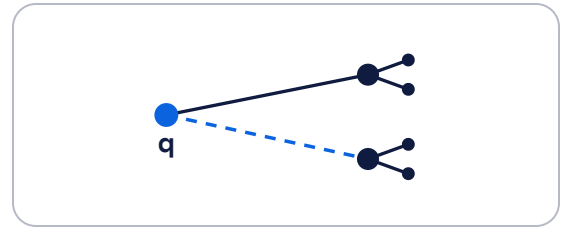
Each recombines the same frozen vectors, at 1.2x to 14.7x the baseline cost.



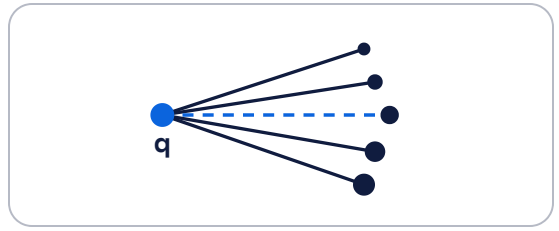

BidirZScore c=1.2



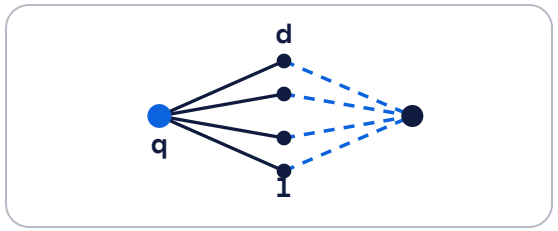

SentMaxSim c=2.2



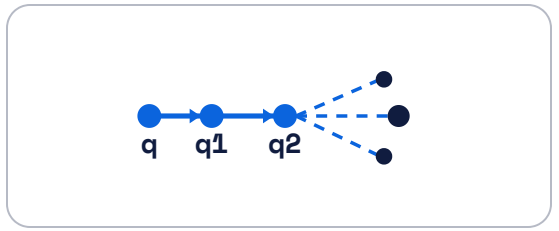

AdaptGranularity c=2.7



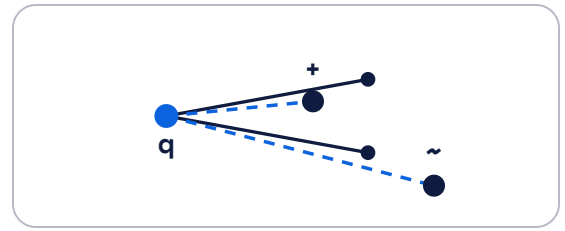

CoverageTriple c=3.7



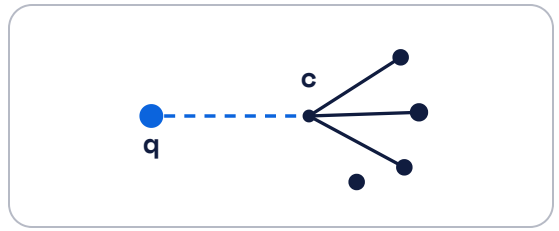

LexicalHybridRRF c=3.9



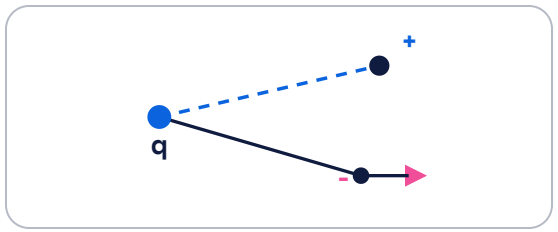

CrossRoundRRF c=3.9



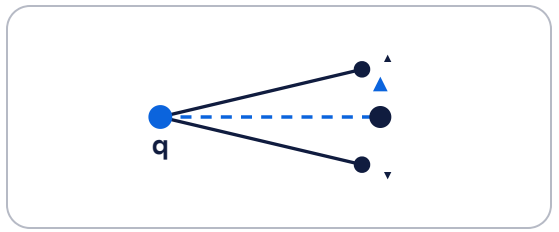

DiverseDualCtx c=5.6



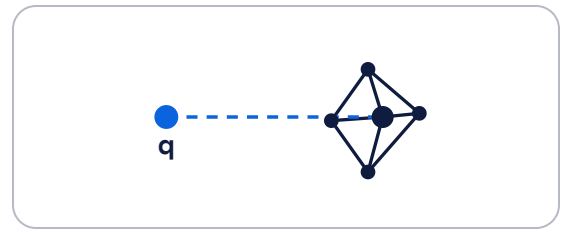

ConsensusRocchio c=6.4



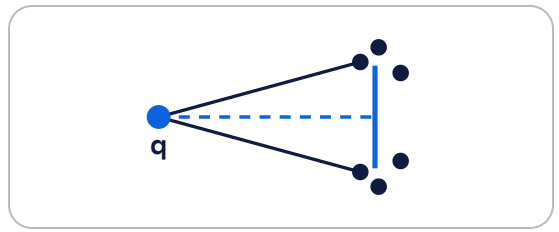

NegContrastive c=7.2




MomentumProg c=9.8

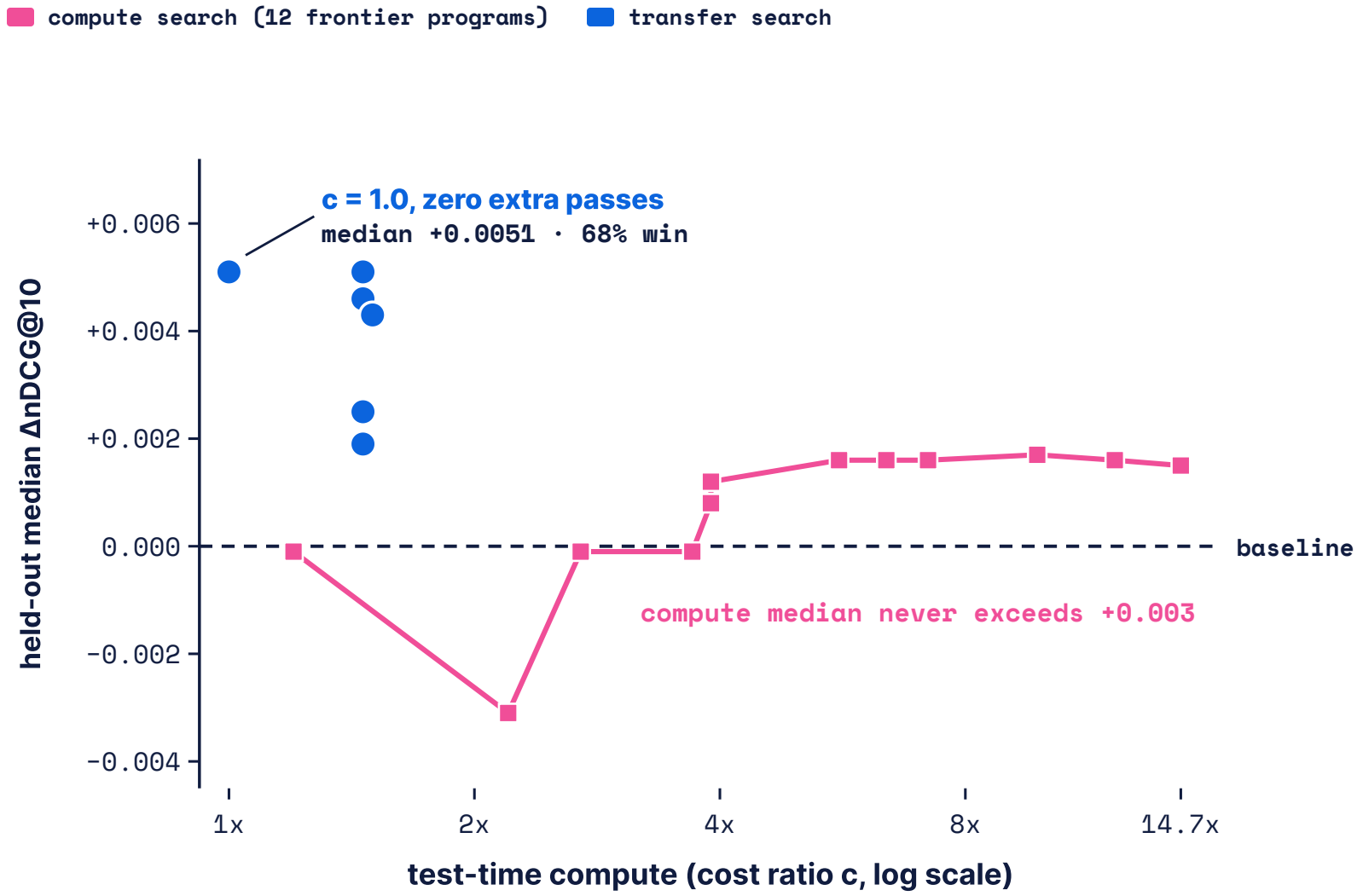



GraphCentrality c=12.2




FisherStability c=14.7

# On held-out tasks, a program at c = 1.0 beats all 12 compute programs




**+0.003**

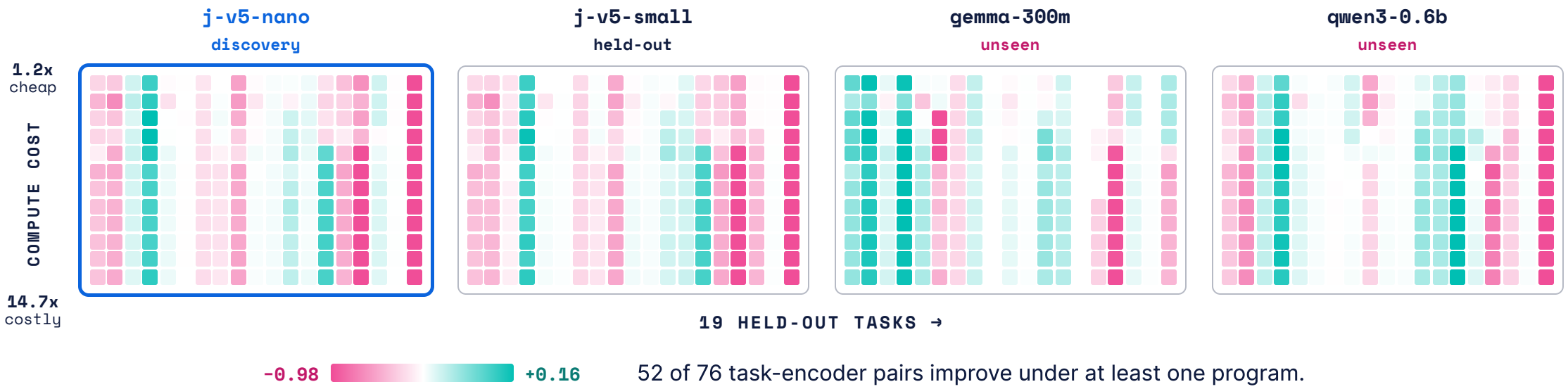
highest held-out median of the compute frontier, up to 14.7x

**+0.0051**

held-out median at c = 1.0, with a 68% win rate

# Compute improves about half the cells and badly hurts others

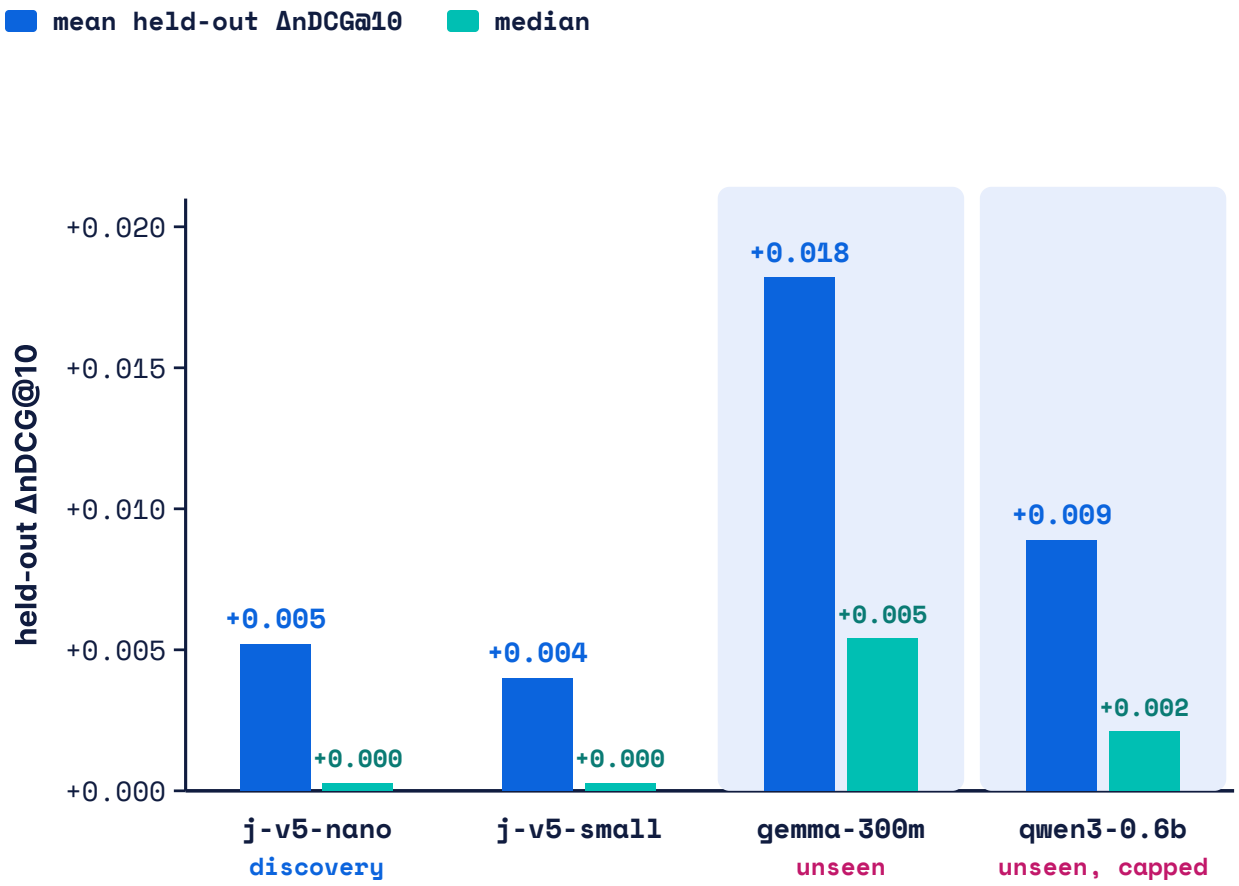
12 programs × 19 held-out tasks × 4 encoders. 485 of 912 cells improve, the worst falls to −0.98, and the pooled mean is −0.016.




52 of 76 task-encoder pairs improve under at least one program.

# A transfer search finds cheap programs that generalize

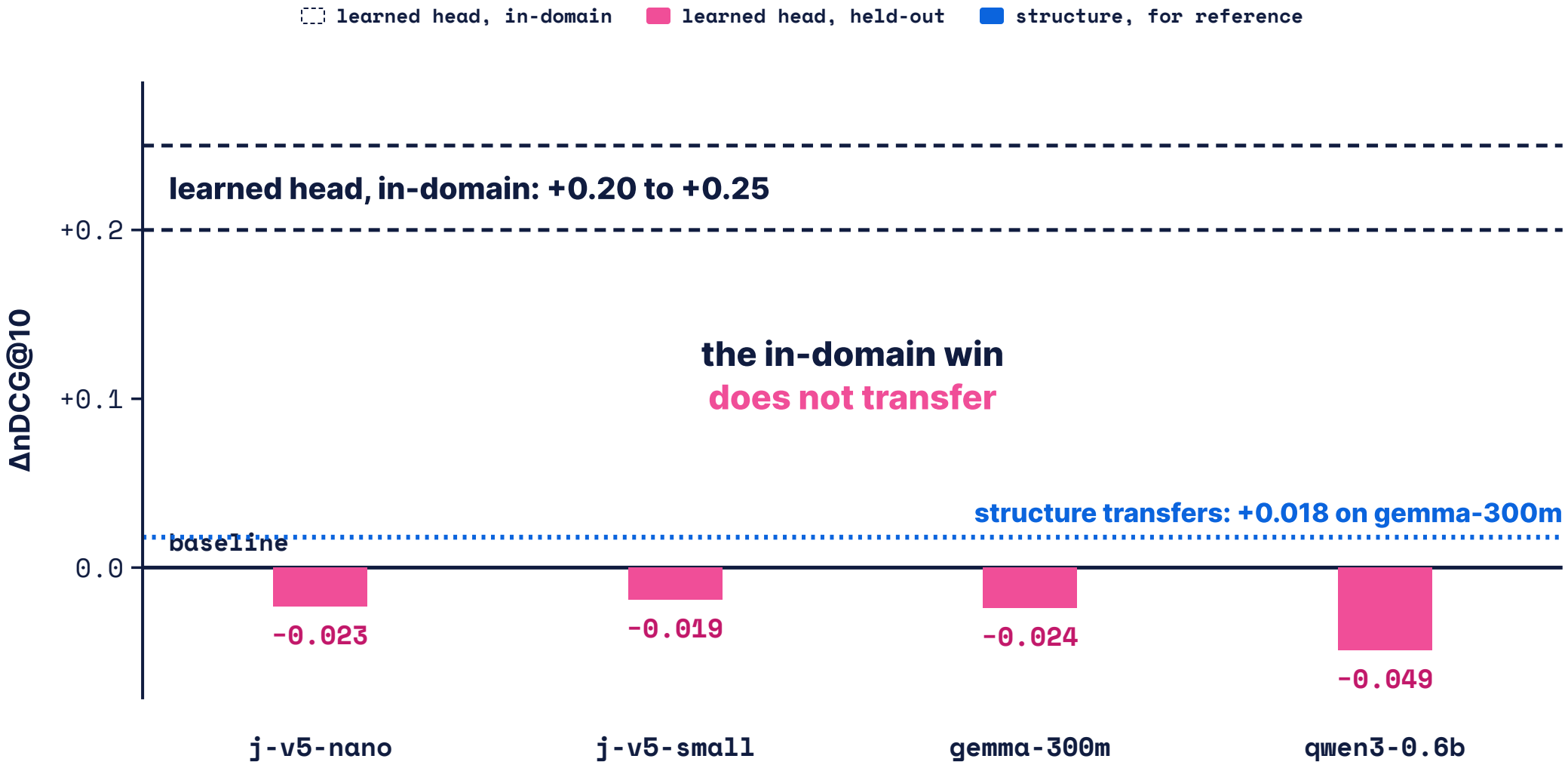
A program is kept only if a held-out validation split improves and no task drops by more than 0.05. Six pass, all at 1.5x or less.




- Mean gains are positive on all four encoders and largest on unseen families.
- Transpose-Consensus never loses a task: **83% win rate**, worst cell **0.000**.
- Applied unmodified to French and Greek: median **+0.016**, **86%** win rate.

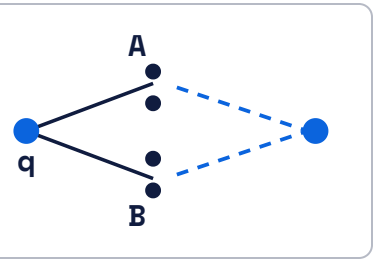
# A learned head at the same data budget memorizes

Linear, low-rank and MLP heads trained on the 14 discovery tasks fall below baseline on every held-out encoder.



# The search re-derives classical IR in embedding space

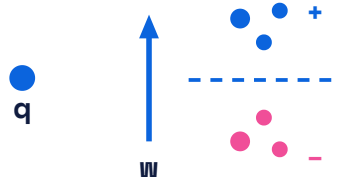
All four rely only on cosine geometry, which every encoder shares.




## Reciprocal Rank Fusion

APPEARED WITHOUT A SEED

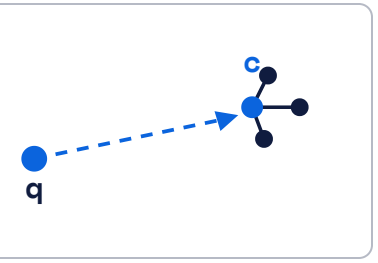
fuse two rankings into one




## Fisher Linear Discriminant

APPEARED WITHOUT A SEED

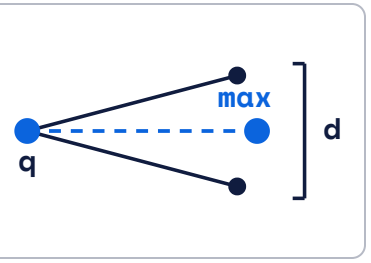
the axis that best separates relevant and non-relevant




## Rocchio Pseudo-Relevance Feedback

OPERATIONALIZED FROM A SEEDED IDEA

pull the query toward the relevant centroid




## Sentence-level MaxSim

OPERATIONALIZED FROM A SEEDED IDEA

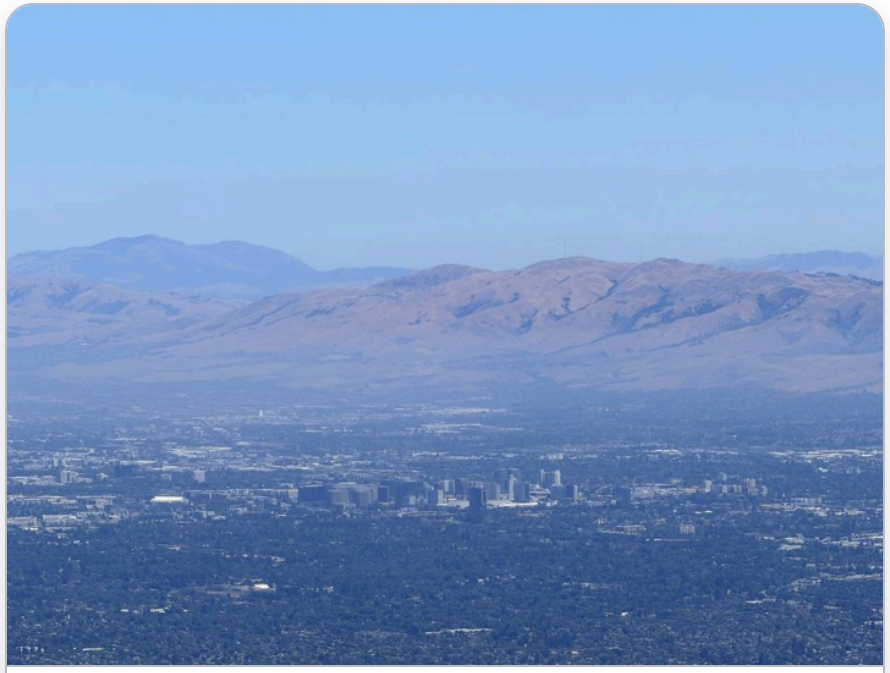
score a document by its best sentence

02

# A retrieval model tags images

# Open-vocabulary tags from a frozen embedding model

The tags are verbatim, with no training, dictionary or second model.



mountain · phoenix · sky · rooftop · location



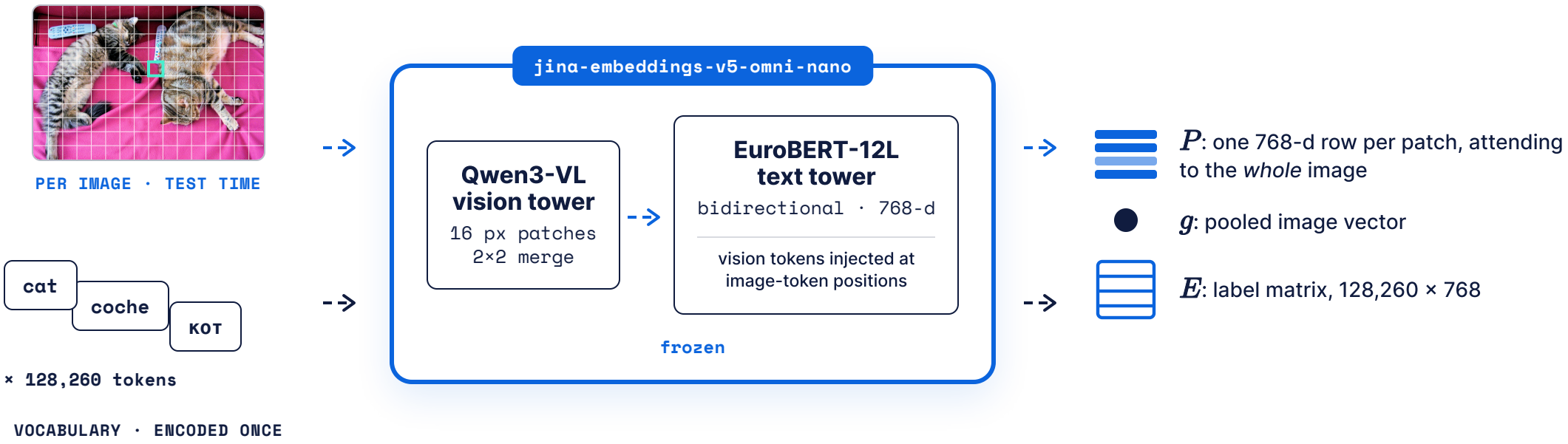



clock · thermostat · dial · dst · wifi

9 photos · top-5 tags each

# Images and text share one embedding space

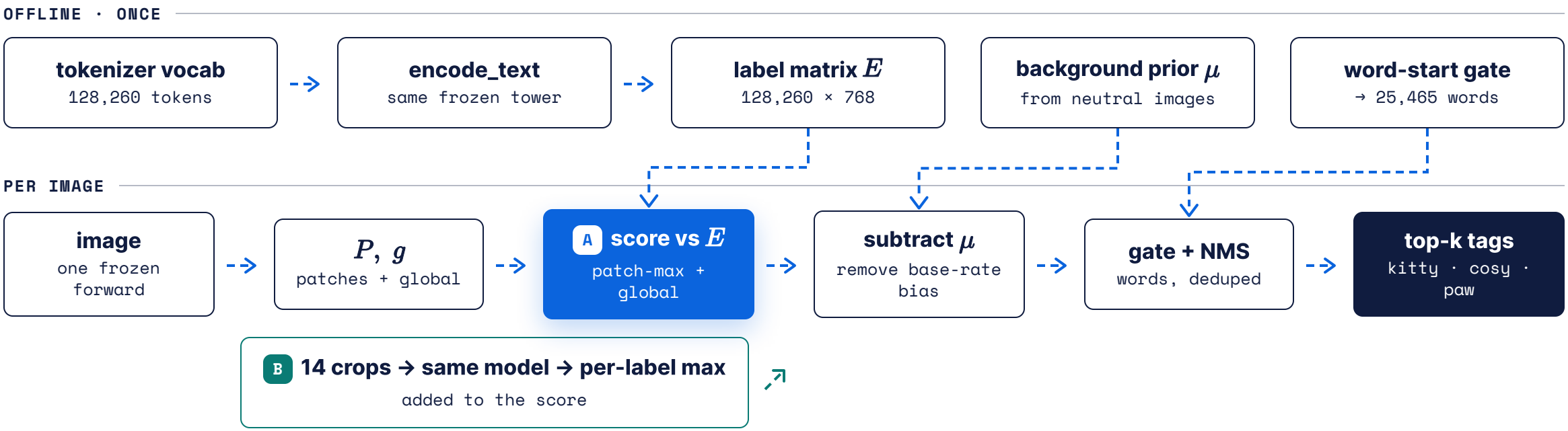
jina-embeddings-v5-omni-nano is a retrieval encoder of about 1B parameters with no tagging head.




Any image patch can be scored against any word in the vocabulary.

# The whole tagger is test-time compute

The label set is the tokenizer vocabulary: 128,260 tokens, gated to 25,465 whole words.

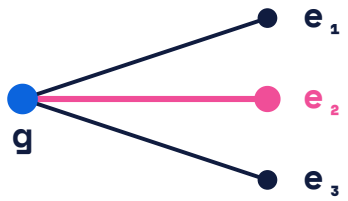



Every box is the frozen model or arithmetic. A reads more of one pass, B adds passes on crops.

# Scoring every patch lifts mAP from 0.264 to 0.635

The patch vectors already exist in the forward pass. Each label keeps its best patch, as in ColBERT.

### GLOBAL POOLED

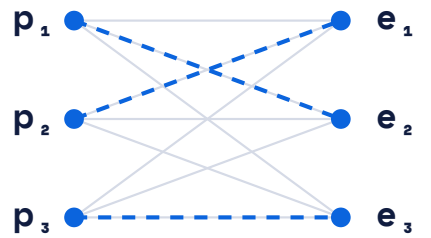



$$s_\ell = \langle g, e_\ell \rangle$$

one vector, dominated by the most salient object

mAP 0.264

### PER-LABEL MAX OVER PATCHES



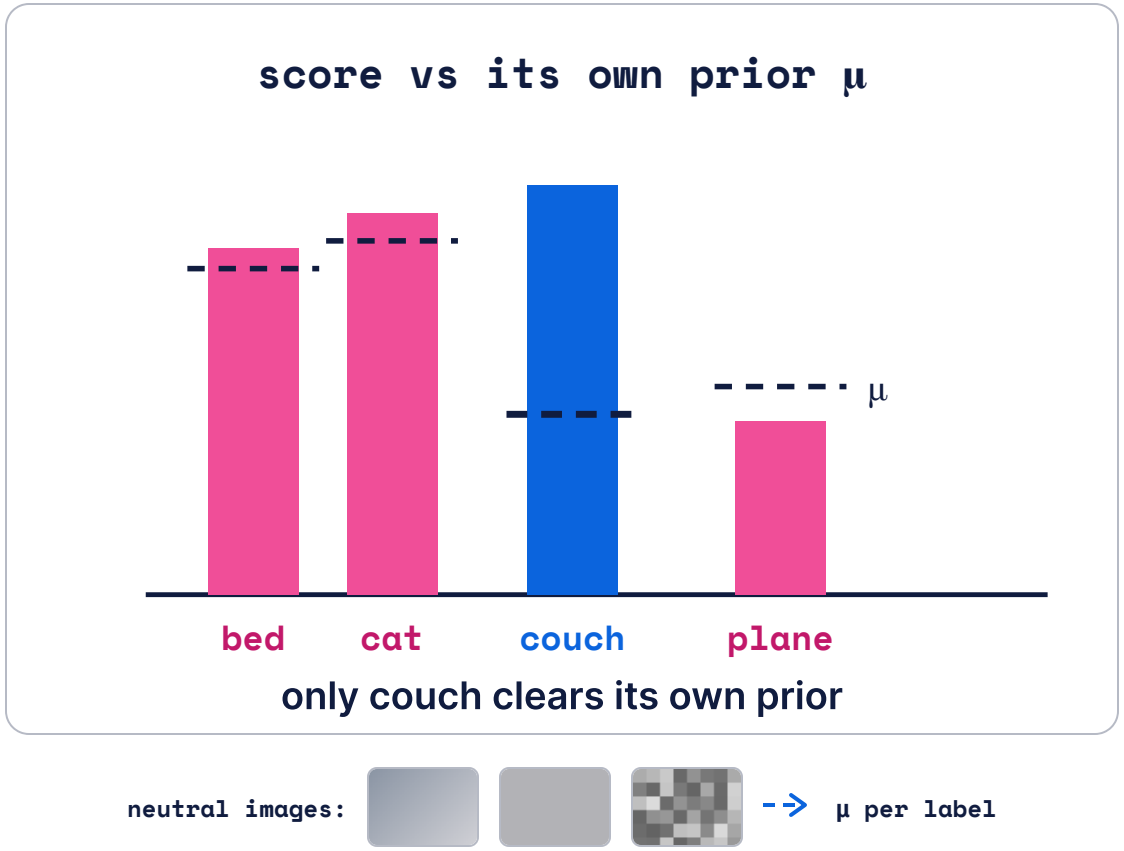

$$s_\ell = 0.7 \max_{p \in P} \langle p, e_\ell \rangle + 0.3 \, \langle g, e_\ell \rangle$$

best patch per label, plus the global vector

mAP 0.635

# Each label is compared with its own prior

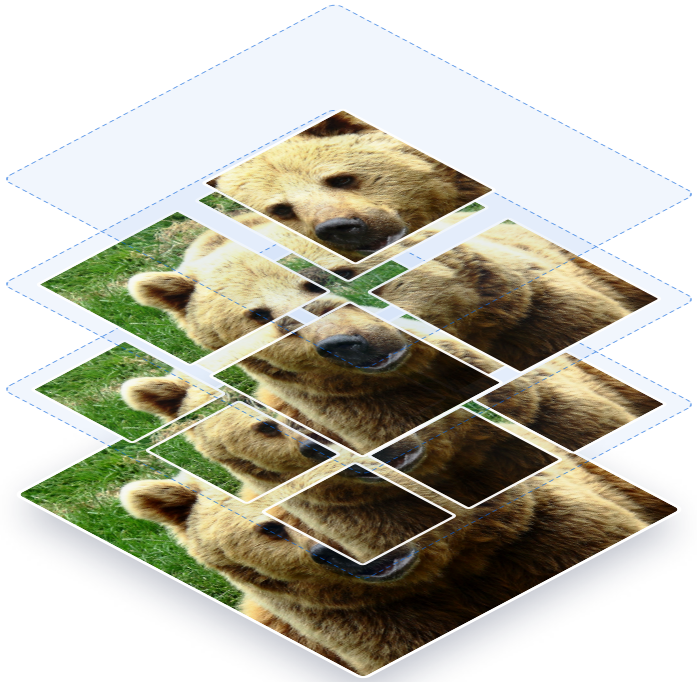
Generic words such as bed and cat score high on every image. A prior from neutral images removes that bias.




- One offline pass, with no calibration set or labels.
- **Word-start gate:** keep tokens that the tokenizer marks as word starts.
- **Embedding-NMS:** drop any word within cosine 0.6 of one already kept, so cat, kitty and кот merge.

# Crops give the model new pixels

A small object fills its own crop, so weak evidence becomes strong. Each label takes its max over 14 crops.

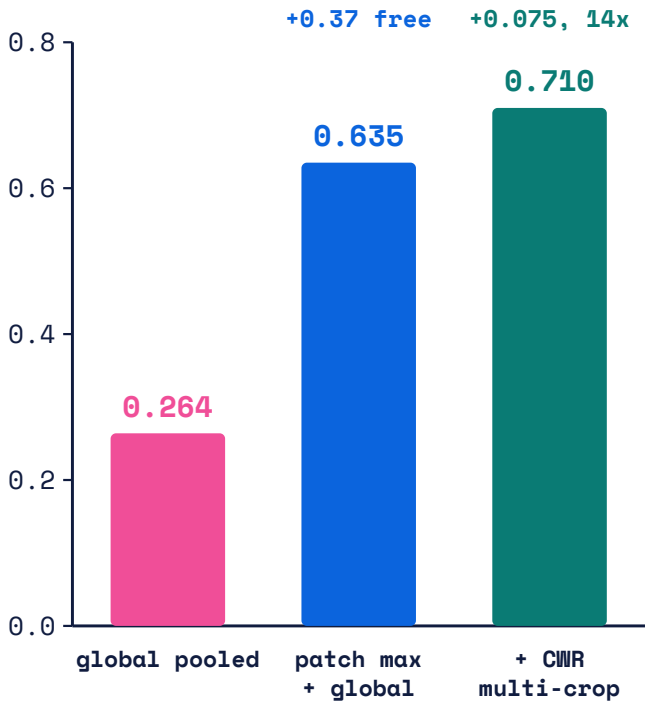


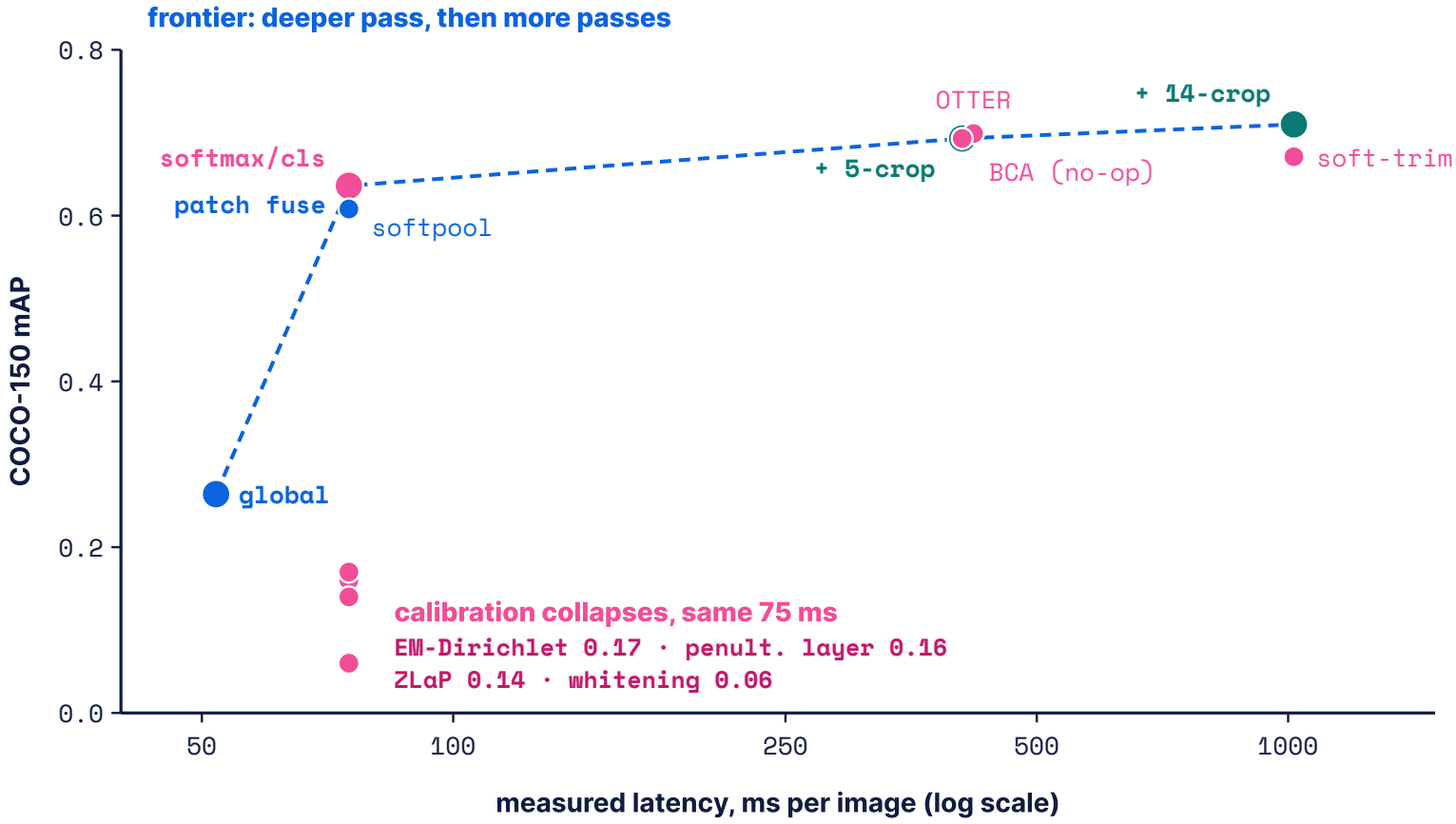

| | | | |
|---|---|---|---|
| 1 pass | wolf | roar | muzzle |
| + 14 crops · per-label max | fur | bear | muzzle |

- **+0.075 mAP** for 14x the passes: 0.635 to 0.710.
- P@1 goes from **0.753** to **0.813**.
- One pass reads **wolf, roar, muzzle**. With crops it reads **fur, bear, muzzle**.

# Accuracy climbs with each kind of new information

Patches give the big step. Crops give the next one at higher latency.

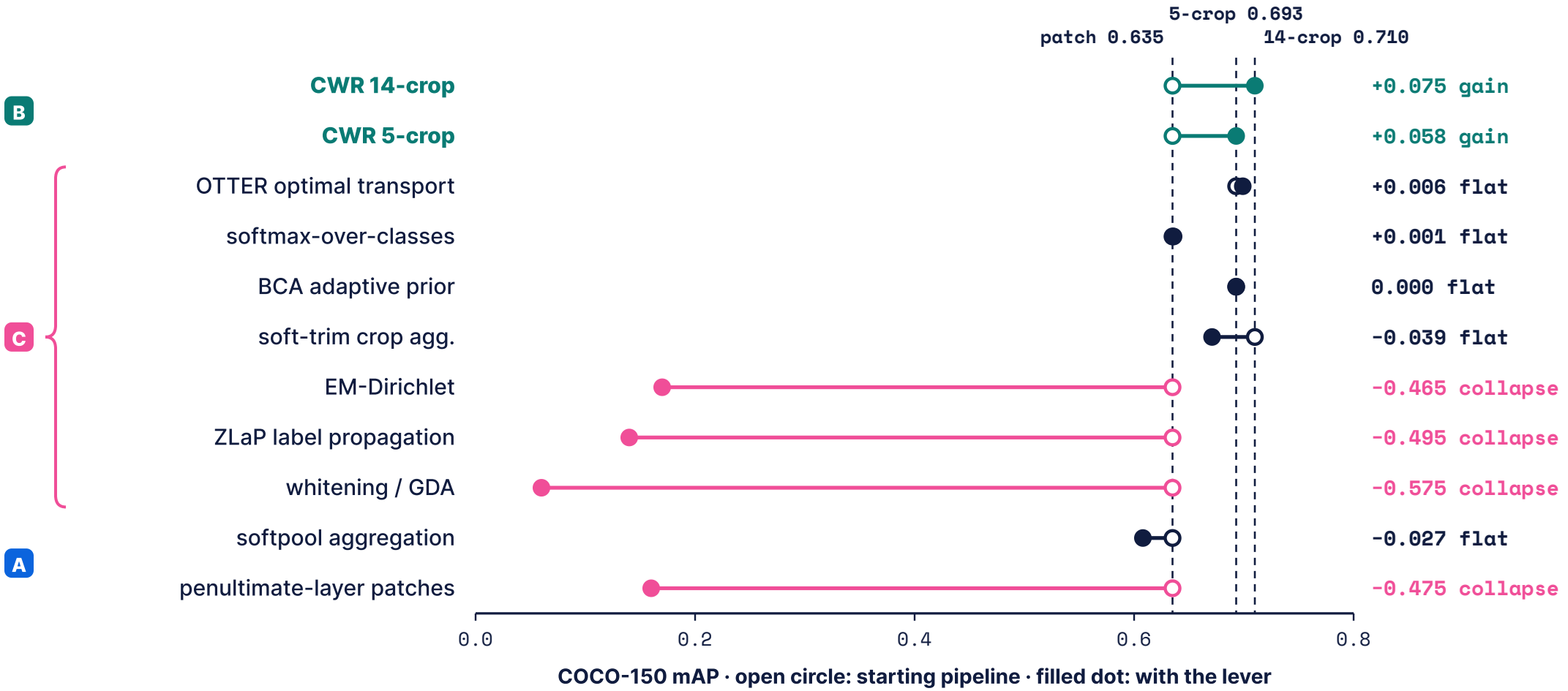



COCO-150 mAP: global → patch → + 14 crops

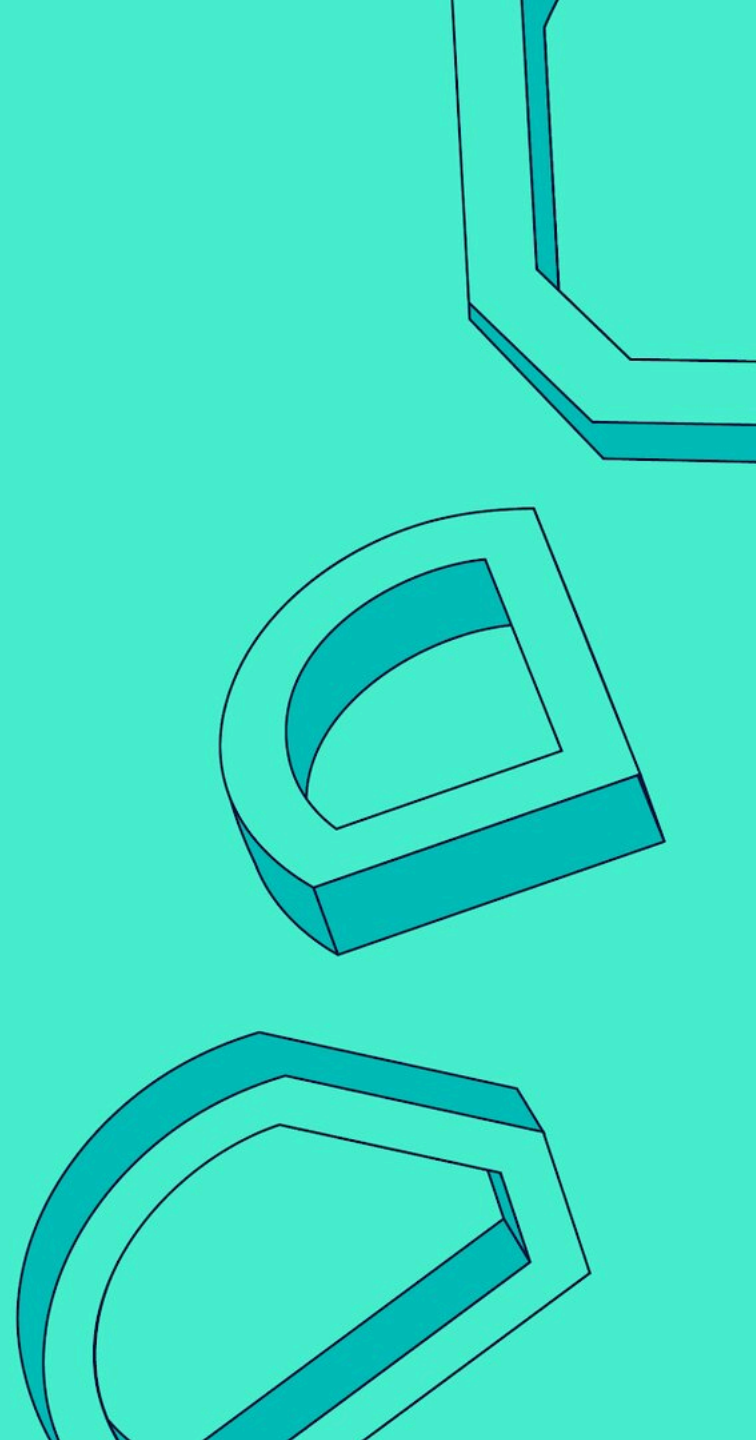
# Calibration does not move accuracy

Recent training-free methods rely on calibration. On this encoder, every calibration lever is flat or collapses.



# Only new information moves accuracy.

Patches and crops add information. Calibration rearranges what is already there.



# Tagging runs on device in Omni

It reuses the patch rows of the embedding pass. Tags become search snippets for images, video and scanned PDFs.

**~0.6 ms**

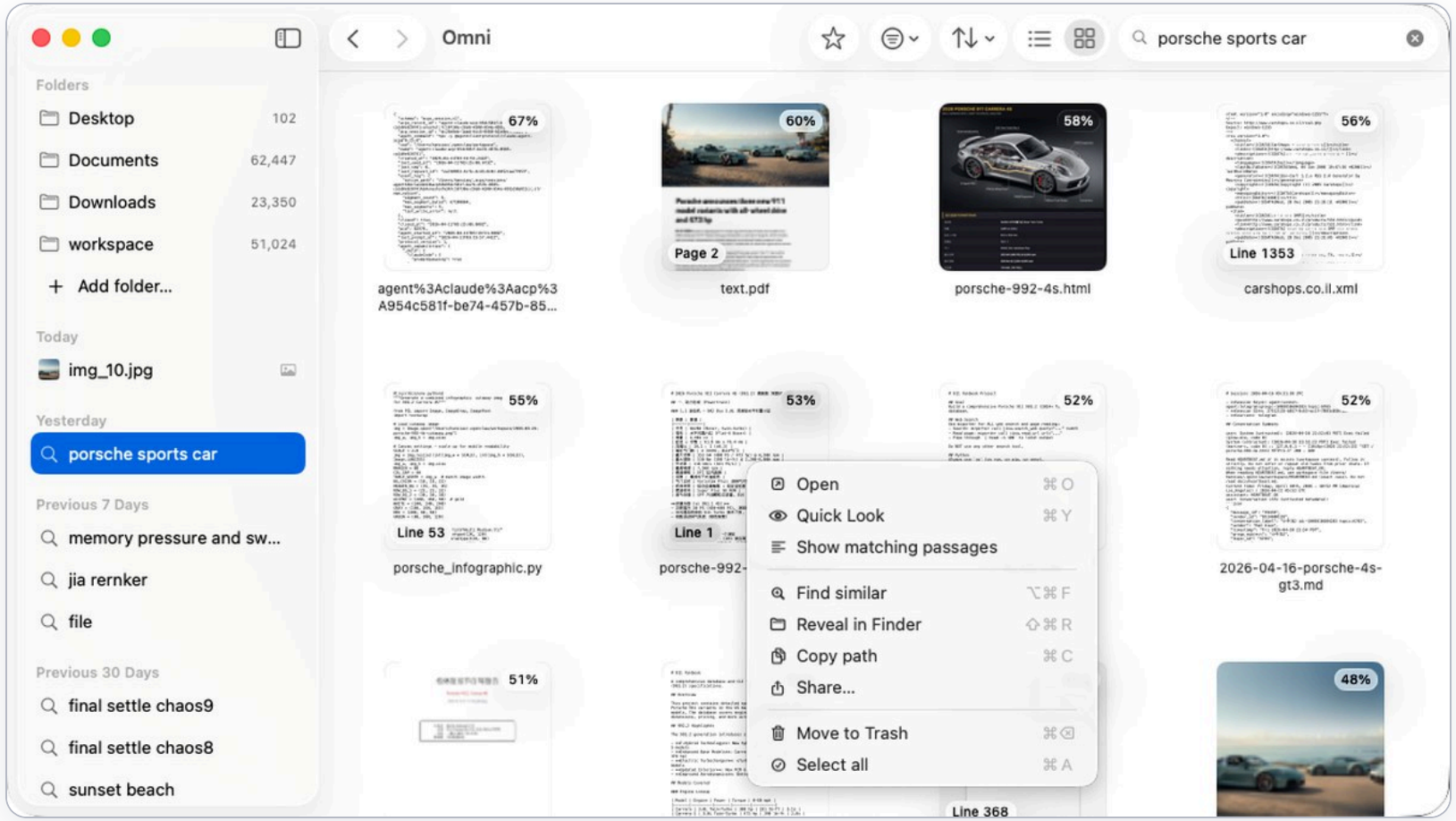
per image

**~4%**

of the embedding step

**64**

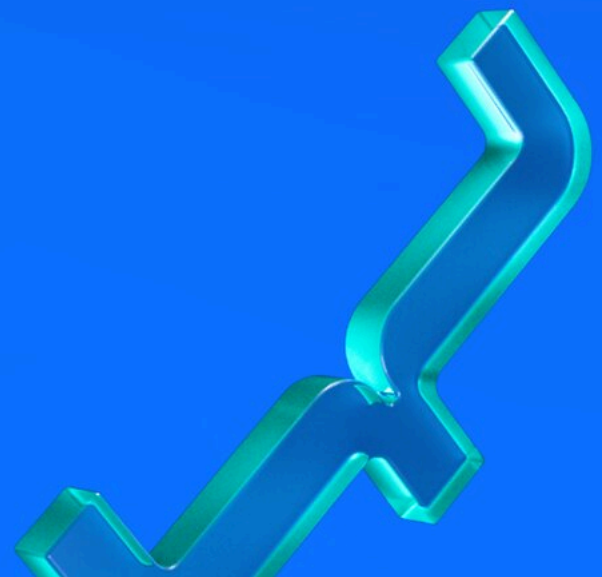
images calibrate the prior



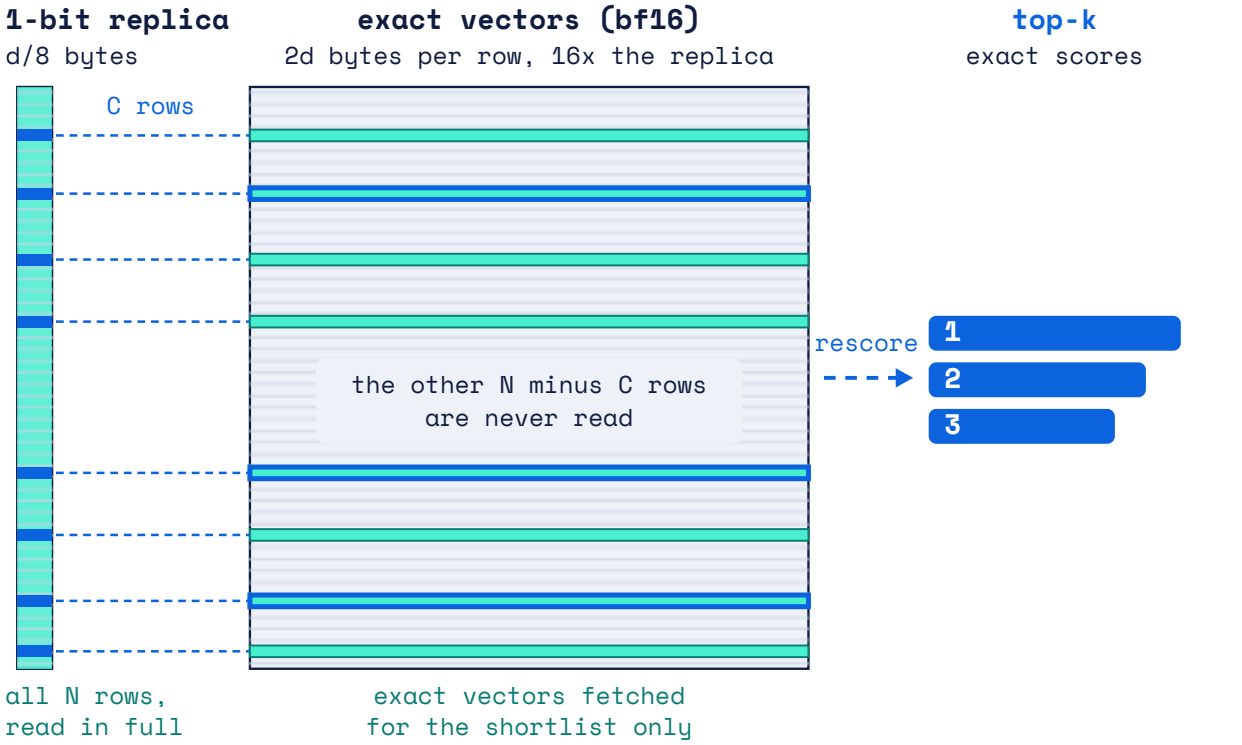


03

# One call replaces an index



# Omni builds no index and spends compute on rescoring




| 01 SCAN | 02 SHORTLIST | 03 RESCORE | 04 TOP-K |
|---|---|---|---|
| every row, 1 bit | top C, coarse | exact, C rows | exact order |

Teal marks what one query reads.

Every query scans the whole corpus. A graph index would add 256 bytes of links per row, **0.95 GB at four million chunks**, close to three times the scanned matrix, and edits keep invalidating it.

**The scan reads a 1-bit replica.** Exact vectors are read only for the shortlist and set the final order.

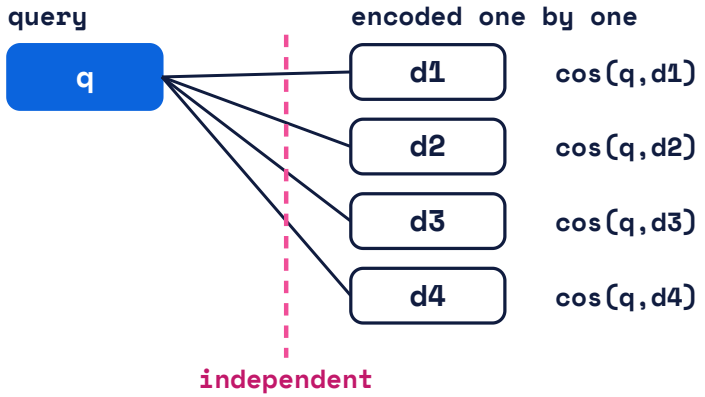
Shortlist size: `min(4096, max(1024, topK×32))`

# Let the candidates see each other in one forward pass

jina-reranker-v3 (0.6B) reads the query and up to 64 documents in one context.

## Bi-encoder

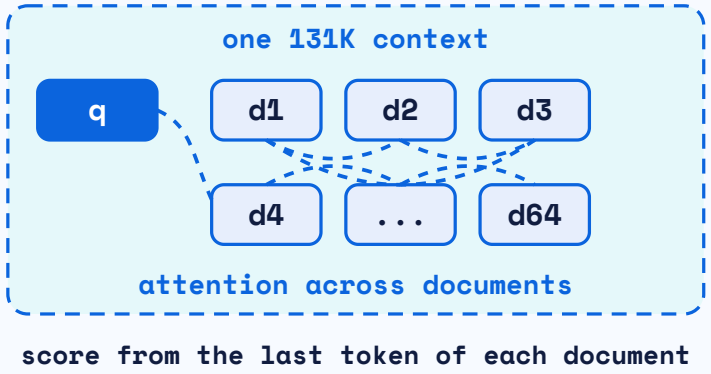
EACH DOCUMENT SCORED ALONE




Each document vector is computed without the others.

## Listwise

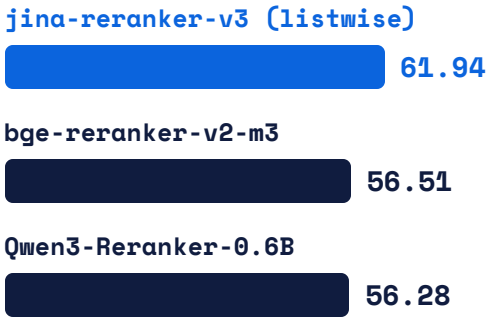
ALL CANDIDATES IN ONE CONTEXT




Each document is scored after attending to every other candidate.

## BEIR nDCG@10

0.6B CLASS



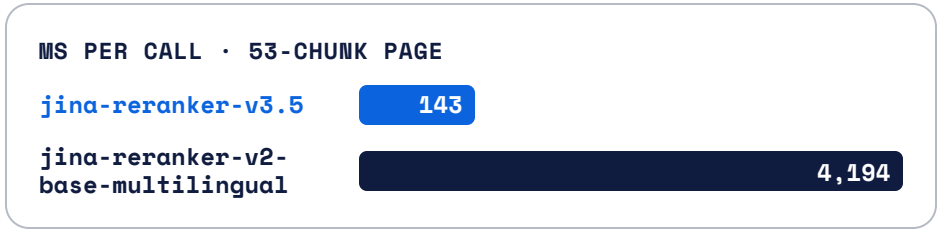

At the same size, the listwise model leads.

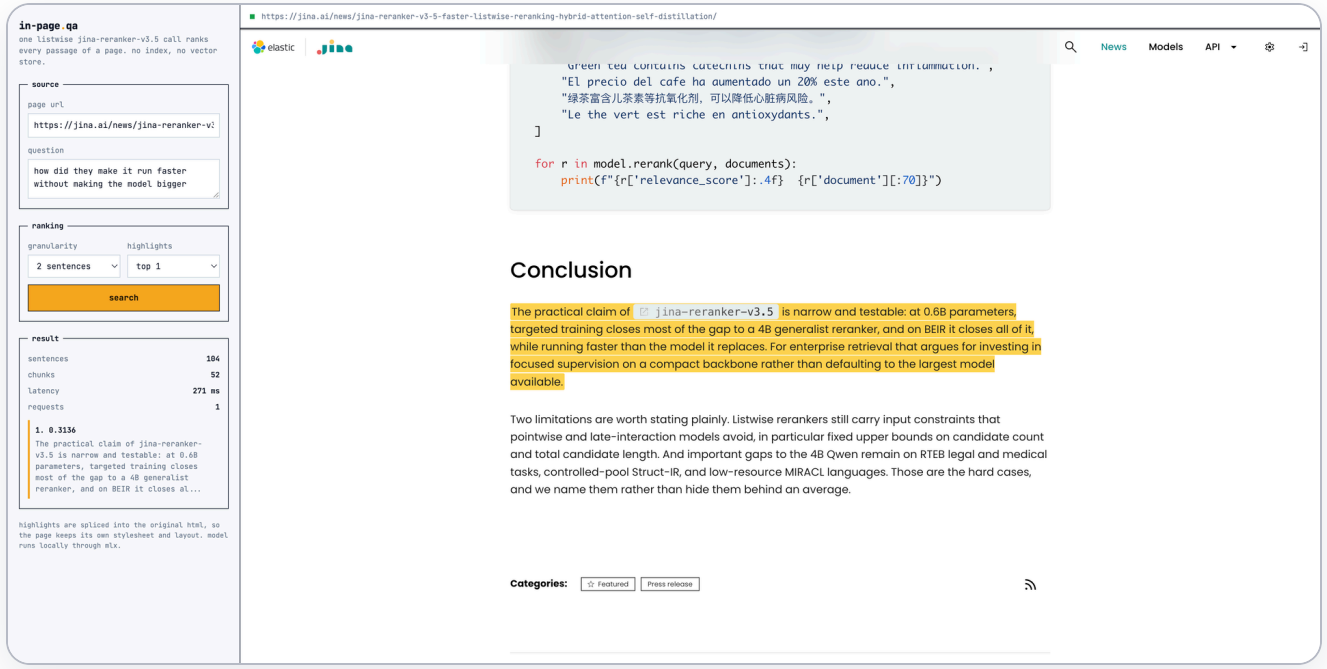
# A single page needs no index

An index pays off over many queries. A page lives for one visit.

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- jina-reranker-v3.5 scores a 53-chunk page in **143 ms** per call.
- Answers are highlighted in place on the original page.



04

# An agent builds the pipeline

# Test-time compute is the pattern for long-horizon agents

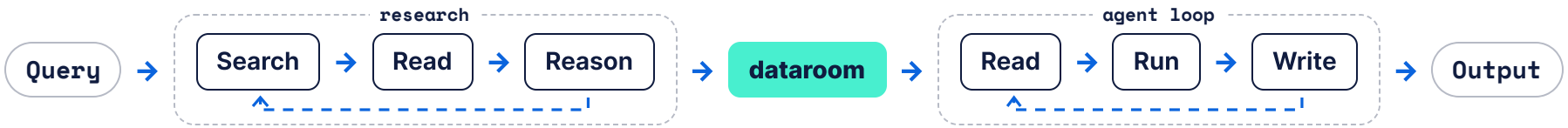
Cheap local tokens build the corpus. Frontier tokens go to the work that needs them.

## 2025 · Deep research




one loop · web IO throughout

## 2026 · Long-horizon tasks

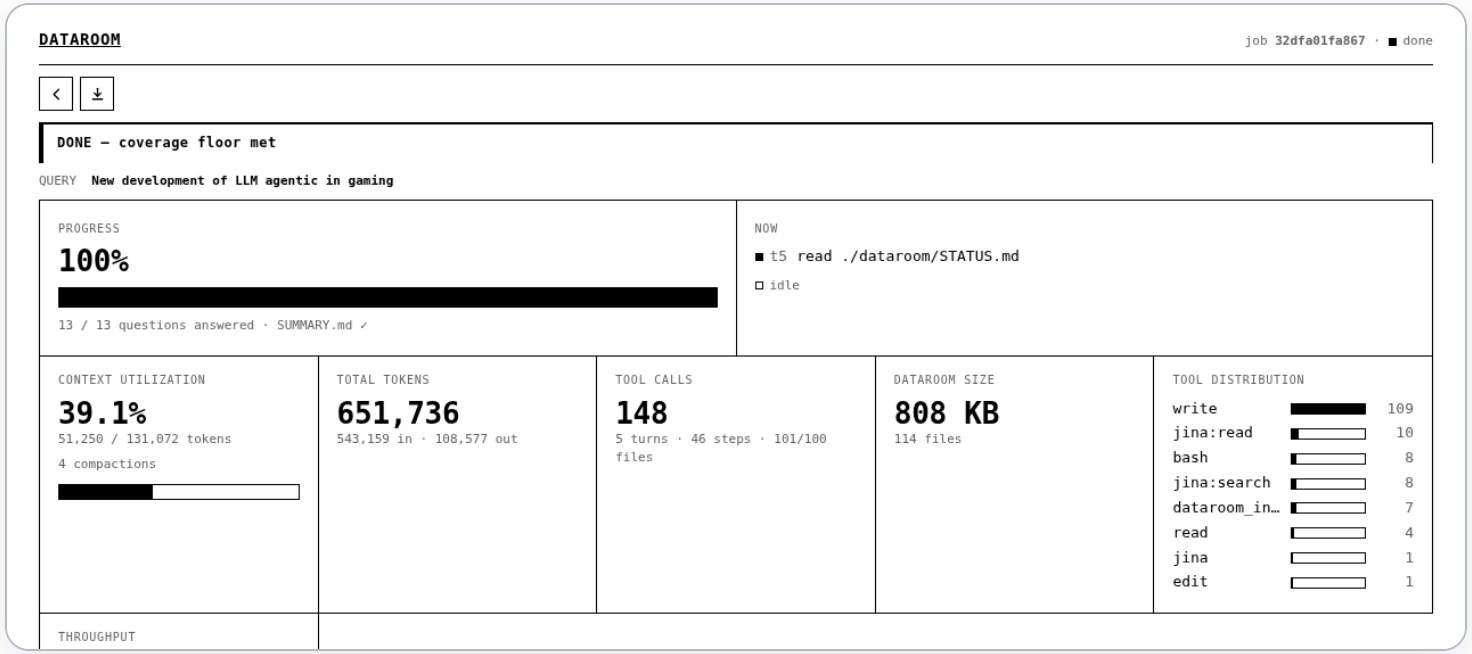



research with web IO → corpus → offline agent loop, no web IO

# dataroom: a token budget builds a corpus

A local model searches, reads and writes until the knowledge sits in one cited .zip.

GITHUB

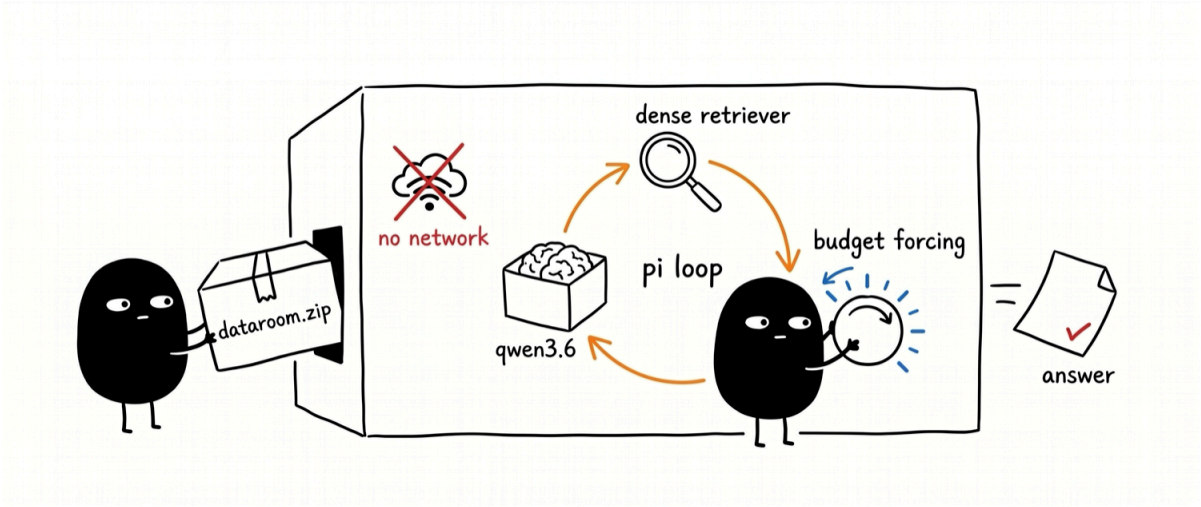



- It stops at a measured coverage floor.
- The job page tracks coverage, tokens and tool calls.
- The .zip goes to searchbox or to a frontier model.

# searchbox: an agent builds its pipeline for each query

The agent gets one dataroom, local tools and no web access.

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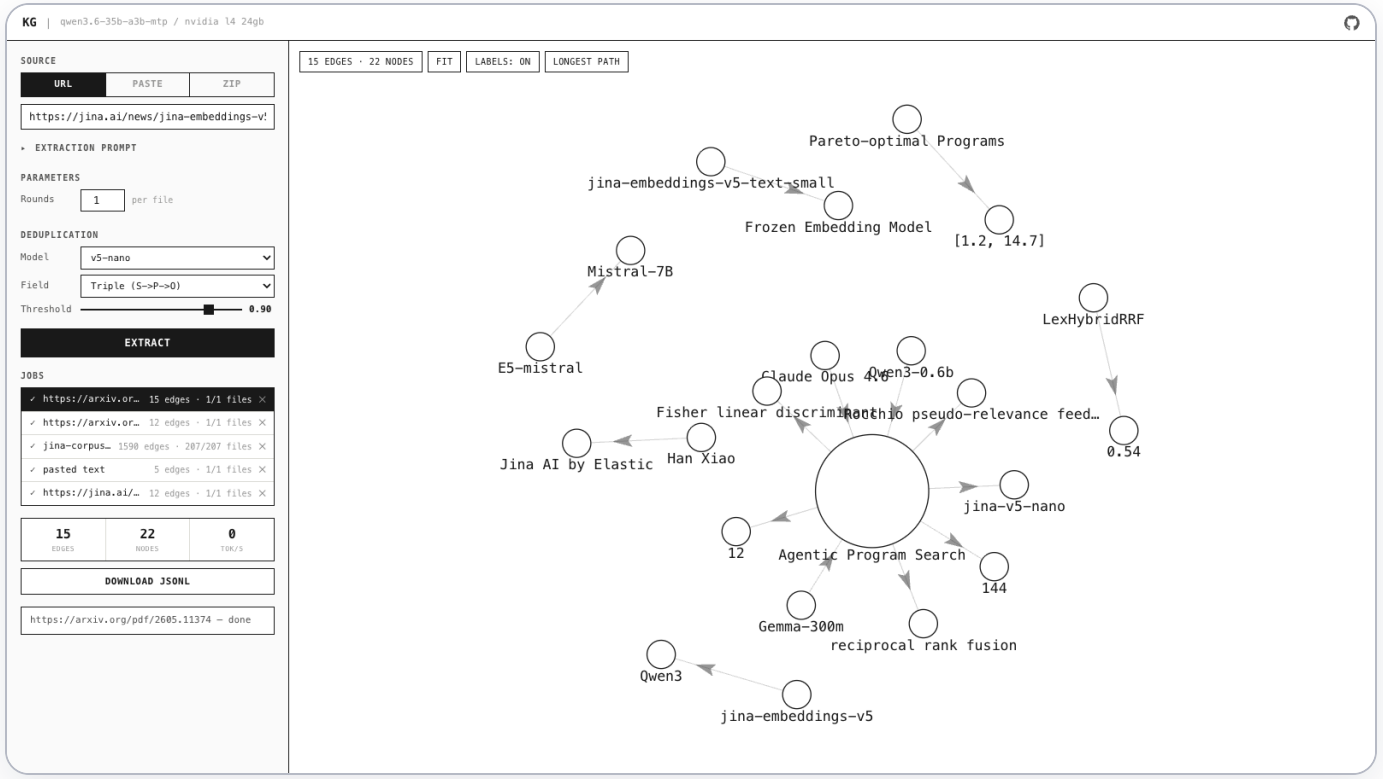



- **Tools:** grep, embed, rerank, similarity, cluster, select_diverse.
- → Which tool does the agent call first?
- → When does grep match a dense retriever?
- → Does a larger token budget help on the hard questions?

# knowledge-graph: hard questions as a private verifier

If one grep finds the answer, every method scores the same. Agentic search needs questions that force several hops.

GITHUB




- Turn the corpus into subject-predicate-object edges.
- Walk paths of length 2 and 3, hide the bridge entities and ask for the endpoint.
- Keep a question only if a closed-book probe fails and a retrieval agent succeeds.

# Lessons from the four demos

## Training-free

Programs, tags, listwise ranking and agent pipelines all run on frozen weights.

## Compute buys quality

Programs gain +0.24 in search, a retrieval model tags images, and one call replaces an index.

## New information drives gains

Patches add +0.37 mAP at no cost, crops add +0.075 at 14x, and calibration adds about zero.

## Check out of domain

+0.24 in search became −0.016 on held-out tasks. Cheap programs and private verifiers pass this check.

# Test-time compute in Elasticsearch today

RETRIEVER · RRF

## Fuse rankings

Fuse lexical and dense results with the `rrf` retriever, which the program search rediscovered.

RETRIEVER · TEXT_SIMILARITY_RERANKER

## Rerank the top-k

Rerank the top hits with a listwise model through the `text_similarity_reranker` retriever.

FIELD · SEMANTIC_TEXT

## Score chunks

With `semantic_text` chunking, rank each document by its best chunk.

AGENTS · BUDGET

## Budget the agent

Give agents a token budget, and test them on private multi-hop questions before raising it.

**Search has always been test-time compute. Spend it where new information enters.**



# Thank you

arXiv:2605.11374 · hanxiao.io · x.com/hxiao

Han Xiao, VP of AI, Elastic

THESE SLIDES

hanxiao.io/elasticon-sf-2026

